

Course Level: Undergraduate Open Elective (B. Tech)

BL: 2 Understand

CO: Recall the basics of food groups, food composition, nutritive value and functions of food.

Module 2: Macronutrients: Carbohydrates

1. Definition

Carbohydrates = Hydrates of carbon

Major macronutrient in food, essential for energy supply.

2. Classification & Sources

Free Sugars (Monosaccharides & Disaccharides)

Monosaccharides: simplest units (e.g., glucose, fructose).

Glucose: found in grapes, berries, corn, carrots; main blood sugar.

Fructose: in honey, fruits (apple, grapes, jackfruit); metabolized in liver, low GI.

Disaccharides: two monosaccharides joined.

Sucrose (glucose + fructose), Lactose (glucose + galactose), Maltose (glucose + glucose).

Sugar alcohols (polyols): e.g., erythritol, maltitol; fermented in large intestine.

Oligosaccharides (Short Chain Carbohydrates)

Examples: raffinose, stachyose, inulin.

Not digested by enzymes; fermented by gut microflora.

Polysaccharides

Long chains of monosaccharides linked by glycosidic bonds.

Digestibility depends on linkage type.

Types:

Starch: Rapidly digestible (bread, cooked potatoes), Slowly digestible (pasta, raw grains), Resistant starch (legumes, prebiotic).

Non-starch carbohydrates: dietary fibre (cellulose, hemicellulose, lignin, pectins, gums).

3. Dietary Fibre

Definition: Indigestible carbohydrates resistant to hydrolysis.

Sources: plant cell walls (bran, stalks, leaves).

Types:

Dietary fibre (natural complex carbs).

Functional fibre (isolated/synthetic).

Daily Value (DV): 28 g.

Functions: stimulates peristalsis, prevents degenerative diseases, supports gut health.

4. Glycemic Index (GI) & Glycemic Load (GL)

GI: speed of blood glucose rise compared to pure glucose (GI = 100).

GL: considers both GI and carbohydrate amount.

Low GL ≤ 10 → minimal impact.

Medium GL 11–19 → moderate impact.

High GL ≥ 20 → significant impact.

In diabetes, GL is more important than GI.

Fresh fruits preferred over juices.

5. Functions of Carbohydrates

- Energy Source: 1 g carbohydrate = 4 kcal.
- Protein Sparing: adequate carbs prevent protein breakdown for energy.
- Nervous System: glucose is indispensable for brain function.
- Muscle Function: glycogen breakdown provides energy during contraction.
- Liver Role: supports detoxification, regulates protein/fat metabolism.
- Nucleic Acid Synthesis: glucose → ribose for DNA/RNA.
- Fat Conversion: excess carbs stored as fat.
- Gut Health: lactose promotes beneficial bacteria, aids calcium absorption.